

GitHub Actions: PreBuilt Workflow

Read the same workflow you will use for Project 2.

Learning Goals

- Distinguish `run` commands from `uses` actions.
- Explain the supplied Hugo workflow's build and deploy steps.

Use the Supplied Workflow

File under `module3/code/` :

```
3_Github_io/3_Github_io_Hugo_Deployment/workflows/hugo.yml .
```

It installs one specific Hugo version (set by `HUGO_VERSION`), then uses GitHub's Pages actions.

- The snippets here explain that file; copy the complete file for deployment.

run vs. **uses**

Each step in the virtual machine does one of two things:

- **run** : executes a shell command, such as **echo** or **hugo -minify** .
- **uses** : reuses a prebuilt action, which is a ready-made, shareable step (for example, **actions/checkout**).

The YAML script

The YAML script has the format of name/uses or name/run to specify what prebuilt actions or what Linux commands to run.

Hugo Checkout and Install

env:

HUGO_VERSION: 0.167.0

- name: Checkout
uses: actions/checkout@v7

- name: Install Hugo

run: |

curl -sfL -o "\${{ runner.temp }}/hugo.tar.gz"

"https://github.com/gohugoio/hugo/releases/download/v\${HUGO_VERSION}/hugo_\${HUGO_VERSION}_linux-amd64.tar.gz"

mkdir -p "\${HOME}/.local/hugo"

tar -C "\${HOME}/.local/hugo" -xf "\${{ runner.temp }}/hugo.tar.gz"

echo "\${HOME}/.local/hugo" >> "\${GITHUB_PATH}"

- `uses` runs the reusable (prebuilt) Checkout action; `run` executes shell commands.
- `@v7` selects that action's major version.
- jThe Install Hugo step downloads the release named by `HUGO_VERSION`, so very run builds with the same Hugo (the `env:` line sits at the top of the file).
- To upgrade Hugo, change that one number.

Configure Pages

```
– name: Setup Pages  
  id: pages  
  uses: actions/configure-pages@v6
```

This step reports the site's base URL for the next step.

- The same workflow works for any username without editing the URL.

Build and Upload

```
- name: Build with Hugo
  run: hugo --minify --baseURL "${{ steps.pages.outputs.base_url }}/"
- name: Upload artifact
  uses: actions/upload-pages-artifact@v5
  with:
    path: ./public
```

The artifact is the built website passed to the deploy job.

Deploy After the Build

Inside `jobs:`, the deploy job is:

```
deploy:
  environment:
    name: github-pages
    url: ${ steps.deployment.outputs.page_url }
  runs-on: ubuntu-latest
  needs: build
  steps:
    - name: Deploy to GitHub Pages
      id: deployment
      uses: actions/deploy-pages@v5
```

`needs: build` waits for the uploaded artifact before deployment.

Permissions and Settings

At workflow level, permissions include:

```
permissions:  
  contents: read  
  pages: write  
  id-token: write
```

In Settings → Pages, choose **GitHub Actions** as the source.

Check Your Understanding

- Which step installs Hugo, and where is its version set?
- Why must the artifact be uploaded before deployment?
- Which step tells Hugo the published URL?

Next: copy the complete workflow into your own site repository.